MIST4550/6550: Energy Informatics

Spring 2026

CRN: 48052, 48055, 67537 & 67538

Instructor Information
Name: John Rios
Office: E315 Ivester Hall
Email: John.Rios1@uga.edu (please include "EI 9:55 AM" or "EI 11:35 AM" in the subject)
Office Hours (online):

- Tuesday and Thursday, 1:00 - 2:00 pm.
- By appointment, in case the regular office hours do not work for you. Please suggest potential dates/times for the meeting.

Find Zoom meeting info on eLC.

Course Delivery
This course meets in person on Monday and Wednesday at:

- CRN 67537 & 67538: 9:55 am - 11:15 am in room 315, Correll Hall
- CRN 48052 & 48055: 11:35 am - 12:55 pm in room 315, Correll Hall

Course Description
Energy Informatics involves analyzing, designing, and implementing systems to increase the efficiency of energy demand and supply networks. This requires collecting and analyzing data to model energy needs and optimize energy distribution and consumption. Students will leverage information systems competencies and multi-disciplinary knowledge to increase societal energy efficiency.

General

Sustainability is usually defined as "meeting the current generation's needs without compromising the ability of future generations to meet their own needs." Sustainability is currently one of the most critical issues facing our world. It will continue to be so for decades as it will take a long time to reverse some adverse environmental changes. Information Systems (IS) and Information Technology (IT) play a critical role in sustainable development, and Energy Informatics is an approach to addressing sustainability by reducing energy consumption. By learning about Energy Informatics, students enrolled in this class will leverage the necessary computing competencies and multi-disciplinary knowledge to contribute to creating a sustainable future.

The teaching approach will blend interactive lectures, discussions, presentations, case studies, and in-class activities.

Changes to the Syllabus/Course
The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

Updates, changes, or clarifications to the syllabus, assignments, or other course-related matters will typically be addressed at the beginning or end of class. I also occasionally post updates to the eLC website and send notifications via eLC's email function or to your official UGA email address. You are responsible for updates via any of these mechanisms.

Prerequisites, co-requisites

You must be familiar with and competent in using personal computers, software productivity packages (word processor, presentation software, spreadsheet), and the Web. The instructor will assume these prerequisites are met since you are registered in this course.

Objectives

Upon successful completion of this course, you will be able to:

- Identify management issues related to designing and implementing energy management systems.
- Apply the Energy Informatics framework to an energy system and model its information elements.
- Leverage information systems to analyze energy data and improve the efficiency of energy systems.
- Explain why the world needs an information strategy for addressing environmental challenges.
- Apply optimization techniques to an energy distribution or consumption problem.

Topics

1. The energy industry
2. Dominant logic
3. Sustainability
4. Energy Informatics
5. Data visualization
6. Optimization methods
7. Spatial-temporal analytics
8. Green IS and Green IT

References

- Watson, R. T., & Boudreau, M.-C. (2011). *Energy Informatics*. Athens, GA: Green ePress. (Available in Kindle format from Amazon)
- GreenBiz newsletter (free subscription recommended)
- eLC course material.

General course policies

- Attendance to all sessions is not mandatory for this course but highly encouraged. Notice that Class Participation is an important component of your course grade. Additionally, a relevant PNAS study shows that "*being actively engaged in the classroom*" is a significant predictor of actual learning.

- Make-up exams may only be given for valid and documented reasons at the instructor's discretion. An unexcused absence from an exam will lead to zero points earned on that exam.
- Backup Policy: Always keep a copy of all material turned in for grading.

File Security

You are responsible for the security of your files. To avoid problems, I recommend regularly making backup copies of your work. In addition, when modeling or visualizing data in R, save your files to prevent data loss in the case of a computer crash.

Active Learning Statement

This is an exciting field that is continually evolving, and we are all here to learn, including the instructor. We will learn from the course material, your practice, classmates, guest speakers, and the news, and we will achieve this goal through a flipped classroom and Active Learning's dynamic interactions and conscious reflections.

UGA Student Honor Code

"*I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others.*" A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at honesty.uga.edu.

UGA Well-being Resources:

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: well-being.uga.edu
- Student Care and Outreach: sco.uga.edu
- University Health Center: healthcenter.uga.edu
- Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/ Fontaine Center: healthpromotion.uga.edu
- Accessibility & Testing: https://accessibility.uga.edu/

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

Inclusive Excellence
The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

Teamwork
In this course, you will work in groups. Each team will have 4-5 members, and a graduate/honors student will be the team leader. The same teams will be used for all group assignments. Additionally, the R assignments will be submitted in pairs.

Review a short report on team effectiveness and establish a team agreement for use by the team. Occasionally someone in a team is not prepared to do their share. In my courses, I recommend that the team gives the freeloader one warning and then fires that person. That person will then do group assignments individually or find a new team to join. The instructor should be notified of changes in team composition immediately.

Professional Development (MIS Undergraduate Only)
The course has the following professional development requirements to help you prepare for the job market and long-term professional success:

- Two events from the following: MIS Department events (e.g., MIS Welcome Back Social) or MIS Department student organization events (e.g., SMIS, WiT). You must register for the event a minimum of 24 hours in advance.

OR

- Take an MIS-related online course that furthers your knowledge. For example, you might take a Python course online, and successfully complete the course exam, possibly receiving a certificate. Get approval from your instructor before you take the course. A course that is shown to take one hour counts for one event, two hours for two events, and so on.

- Attend events or complete activities that are consistent with the course that you are taking. For example, if you are taking MIST 5620, attend a Society of Business Intelligence (SBI) presentation.

- There is no "double counting" of attendance at events. Each event attended can only count for a single course. Within MISTERS, select for which course you want your attendance to count.

- If you have potential conflicts with meeting the professional development requirements or think there are better development activities for your situation, meet with your instructor to discuss the possibilities. This meeting must be at the start, not the end of the semester and is your responsibility to schedule. Requests for alternative ways of meeting the professional development requirements must be entered into MISTERS by the end of 13th week.

- You may want to include the following on your resume: "I have completed XXX hours of professional development, including attending multiple professional meetings, networking with industry professionals, and completing LinkedIn Learning courses."

Grading and Deliverables

	Item	Points	Graduate/honor points
Group	Group Assignments	18	18
	Graduate/honors project	-	8
	Business Case Presentation	4	4
Pair	Programming Assignments	20	20
Individual	G0 - Climate Solutions 101	3	3
	Introduction to Programming	3	3
	Class participation	10	10
	Tech Talks	5	5
	Midterm I	15	15
	Midterm II	15	15
	Course evaluation completion	1	1
	Professional Development	2	-
	Peer Evaluation	4	4
Total **		100	106*

* Will be adjusted to a score out of 100

** Your final points score for the class is based on the weights in the preceding table. Do not use eLC's computed score.

There are at least four Group Assignments and four Programming Assignments in this course. Late assignments will incur a deduction of 10% for each day. Assignments **submitted later than three days from their original due date will not be accepted**.

If you cannot complete an exercise on time or take an exam at the specified time, please advise the instructor as soon as possible so that alternative arrangements can be made.

There is an **optional cumulative final exam** for the course. If you intend to take the optional final exam, you must inform the instructor of this via email no later than **5:00 pm, April 24th, 2026**. *If you take the cumulative final exam, then the midterm exams will count for 10% each, and the final exam will count for 10% of your final grade.*

Use of AI for Coursework

Unless explicitly stated, artificial intelligence-based technologies, such as ChatGPT, must not be used to generate responses for student assignments. Unauthorized use of artificial intelligence software or word-mixing software to write your paper or disguise plagiarized work is considered unauthorized assistance in this course.

Grade Ranges
Course grades are assigned using the following ranges:

- A = 93 or above
- A- = 90 to 92.99
- B+ = 87 to 89.99
- B = 83 to 86.99
- B- = 80 to 82.99
- C+ = 77 to 79.99
- C = 73 to 76.99
- C- = 70 to 72.99
- D = 60 to 69.99
- F = 59.99 or below

Grades will not be rounded up. For example, an 89.99 is a B+ and will not be changed to a 90.

Schedule

Date	Topic	Assignment due date @ 11:59pm the day before
Mon 1-12	Syllabus and Introduction *The problems* How global warming works (CC) The secret to rising sea levels - thermal expansion (CC) Ocean acidification by the alliance for climate education (CC) Can climate change cause war? (CC) Why is biodiversity important? *Some solutions* Climate Solutions 101 - Project Drawdown	More about the problem and solutions The changing state of the climate (CC) Rising sea levels (CC) Acid test: The global challenge of ocean acidification (CC) Climate change is happening. Here's how we adapt (CC) Human activities that affect biodiversity (CC) Biodiversity loss Drawdown Georgia -
Wed 1-14	Energy and sustainability	G0
Mon 1-19	**Martin Luther King Jr. Day- No Classes**	
Wed 1-21	Energy and sustainability	
Mon 1-26	Guest Speaker	
Wed 1-28	Technological solutions to climate change	G1
Mon 2-2	Introduction to Programming	P0
Wed 2-4	Energy Informatics - origins and general concepts	
Mon 2-9	Energy Informatics - origins and general concepts	
Wed 2-11	Green IS & Green IT	P1
Mon 2-16	National energy and sustainability policy	G2
Wed 2-18	Carbon Scopes and ESG Data Midterm Review	
Mon 2-23	**Midterm 1**	
Wed 2-25	Capital, systems, and objects - Introduction to Systems Dynamics	
Mon 3-2	Systems dynamics modeling with En-ROADS	G3
Wed 3-4	Digital data streams and digital twinning Spatial-temporal data analytics	
Mon 3-9	**Spring Break - No Classes**	
Wed 3-11	**Spring Break - No Classes**	
Mon 3-16	Energy Time Series Analysis	
Wed 3-18	Energy Time Series Analysis	
Mon 3-23	Energy data visualization	P2
Wed 3-25	Energy data visualization	
Mon 3-30	Introduction to Optimization: Energy Applications	P3
Wed 4-1	Introduction to Optimization: Energy Applications	
Mon 4-6	Smart Grid and Smart Buildings	P4

Wed 4-8	Smart Grid and Smart Buildings Midterm Review	
Mon 4-13	**Midterm 2**	
Wed 4-15	Graduate and Honors Project Presentations	G&H
Mon 4-20	What else can we improve?: UN SDG	G4
Wed 4-22	What else can we improve?: UN SDG	
Mon 4-27	Business Case Presentation Course Wrap-Up	Case Presentations
	Final Exam: Scheduled by The Office of the Registrar	